ICT330

Examination - July Semester 2024

Database Management Systems

Monday, 11 November 2024 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **9** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A company sells items to the public as well as to its members. Currently, a purchase can only be made physically at one of its several branches. Each branch has a unique branch name, e.g., Bedok 2, and its address is displayed alongside the branch name, on the company website.

When a member makes a purchase, he needs to tap his member card to earn points which he can use to offset future purchases. A member has a unique member id. The member name and points earned thus far should also be recorded.

Each purchase has a unique purchase id and a date and time of purchase. In addition, if the purchase is made by a member, the receipt will indicate the member id, points he has before the purchase, the points used to offset the purchase, if used, with the amount to deduct from the purchase for the points used to offset, the points earned for the purchase and the resulting total points accumulated.

The receipt should also indicate the item code and item name of the items purchased, the quantity purchased and the unit price of the items.

Without adding any attribute/identifier not described in the question, **construct** a conceptual entity relationship (ER) model from the statements of requirements to represent the data model, showing the data requirements in one diagram.

State your assumption(s) for only data requirements that are not specified.

Question 1a

Include entities, with entities name, relevant attributes and identifier.

(10 marks)

Question 1b

Include relationships, with maximum and minimum cardinalities and relationship name.

(10 marks)

Question 2

You are given the table:

HamperPurchase(hamperName, hamperPrice, itemCode, itemName, itemDesciption, purchaseId, purchaseQty)

This table is used by a company to record hamper purchases. Each hamper is identified by hamperName, and is sold at one price. For example, Exam Care Hamper is sold at $24.95 and both Tea Break Special Hamper and Tropical Fruit Hamper are sold at $34.95.

Each hamper contains a number of items, each identified by itemCode. An item can be packaged in different hampers. For example, Ferrero Nutella Biscuit can be found in both Exam Care Hamper and Tea Break Special Hamper.

Each purchase is identified by a purchaseId. A purchase may consist of one or more hampers. Each hamper may be purchased at a different purchased quantity. For example, a purchase with purchaseId A03451 may consist of 5 units of Exam Care Hamper and 4 units of Tea Break Special Hamper.

Question 2a

Formulate and list the functional and multi-valued dependencies (MVD). Do not introduce any new columns not in the given table. State your assumption(s) for only data requirements that are not specified.

(5 marks)

Question 2b

Draw dependency graph and propose candidate key(s) for the table. Show composite keys in brackets.

(3 marks)

Question 2c(i)

Explain and discuss whether the MVDs in your answer to Q2(a) are subsumed or not subsumed.

(2 marks)

Question 2c(ii)

Apply the BCNF approach covered in the course text to normalise the table to Boyce Codd Normal Form (BCNF) and Fourth Normal Form (4NF).

- Show how you apply the steps to arrive at the tables.
- Explain whether the structure of each derived table at each iteration is already in BCNF and 4NF (if applicable).

(8 marks)

Question 2d

Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:

TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3

Question 3a

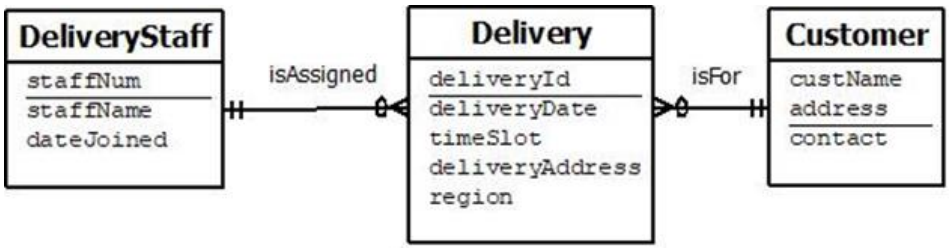

Figure Q3(a)

Question 3a(i)

Transform the ERD in Figure Q3(a) into a logical model.
Show the tables in the logical model in the form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key,
- indicate (FK) against the foreign keys, if any, including integrity constraints in the form:

TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(5 marks)

Question 3a(ii)

Refer to the description of the columns for the column definition, and then code the appropriate SQL statements to **implement ONLY** the **Delivery table** in Figure Q3(a).

All columns are required.

staffNum	Exactly 4 characters in length, 2 alphabets followed by 2 digits.
staffName	At most 60 characters in length
dateJoined	Datetime with default datetime and must be not earlier than the current datetime
deliveryID	An integer, auto generated with starting value 1 and increment 1
deliveryDate	Datetime with default date 1 week from current datetime and must be at least with 1 week from current datetime and not more than 3 weeks later than the current datetime
timeSlot	exactly 9 characters in length, with a value either morning, afternoon or night
deliveryAddress	At most 150 characters in length
region	A number from 1 to 8
custName	At most 60 characters in length
address	At most 150 characters in length
contact	Exactly 9 characters in length, all digits

Constraints for the relationships in Figure Q3(a):

isAssigned	delete and update are disallowed
isFor	delete disallowed and update cascaded

(7 marks)

Question 3b

This question is based on the tables derived from the ERD in Figure Q3(a).
Write a trigger to disallow more than 8 deliveries for any time slot on any given date.
State any assumptions you made in implementing the trigger.

(8 marks)

Question 3c

For this question part, you are allowed to use inline views in your query formulation, but you must NOT create views using the CREATE VIEW statement.

Use the database tables in Figure Q3(c).

ArtistID	LastName	FirstName	Nationality	DateOfBirth	DateDeceased
1	Miro	Joan	Spanish	1893	1983
2	Kandinsky	Wassily	Russian	1866	1944
3	Klee	Paul	German	1879	1940
4	Matisse	Henri	French	1869	1954
5	Chagall	Marc	French	1887	1985
11	Sargent	John Singer	United States	1856	1925
17	Tobey	Mark	United States	1890	1976
18	Horiuchi	Paul	United States	1906	1999
19	Graves	Morris	United States	1920	2001
1001	Tamayo	Rufino	Mexican	1899	1991

Artist table

WorkID	Title	Copy	Medium	Description	ArtistID
570	Untitled Number 1	Unique	Monotype with tempera	4.3 x 6.1 in. Signed	17
571	Yellow Covers Blue	Unique	Oil and collage	71 x 78 in. - Signed	18
578	Mid-Century Hibernation	362/500	High Quality Limited Print	Northwest School Expressionist style	19
580	Forms in Progress I	Unique	Color aquatint	19.3 x 24.4 in. - Signed	17
581	Forms in Progress II	Unique	Color aquatint	19.3 x 24.4 in. - Signed	17
585	The Fiddler	252/1000	High Quality Limited Print	Shows Belarusian folk-life themes and symbology	5
586	Spanish Dancer	588/750	High Quality Limited Print	American Realist style - From work in Spain	11
587	Broadway Boggie	433/500	High Quality Limited Print	Northwest School Abstract Expressionist style	17
588	Universal Field	114/500	High Quality Limited Print	Northwest School Abstract Expressionist style	17
589	Color Floating in Time	487/500	High Quality Limited Print	Northwest School Abstract Expressionist style	18
590	Blue Interior	Unique	Tempera on card	43.9 x 28 in.	17
593	Surf and Bird	Unique	Gouache	26.5 x 29.75 in. - Signed	19
594	Surf and Bird	362/500	High Quality Limited Print	Northwest School Expressionist style	19
595	Surf and Bird	365/500	High Quality Limited Print	Northwest School Expressionist style	19
596	Surf and Bird	366/500	High Quality Limited Print	Northwest School Expressionist style	19

Work table

TransactionID	DateAcquired	AcquisitionPrice	AskingPrice	DateSold	SalesPrice	CustomerID	WorkID
161	2013-06-28	7500.00	15000.00	2013-09-29	13750.00	1033	570
171	2013-08-23	35000.00	60000.00	2013-09-29	55000.00	1000	571
181	2013-10-11	250.00	500.00	NULL	NULL	NULL	578
201	2014-02-28	2000.00	3500.00	2014-04-26	3250.00	1040	580
202	2014-02-28	2000.00	3500.00	2014-04-26	3250.00	1040	581
225	2014-06-08	125.00	250.00	2014-09-27	225.00	1051	585
226	2014-06-08	200.00	400.00	NULL	NULL	NULL	586

227	2014-06-08	250.00	500.00	2014-07-11	475.00	1051	587
228	2014-06-08	250.00	500.00	NULL	NULL	NULL	588
229	2014-06-08	250.00	500.00	NULL	NULL	NULL	589
241	2014-08-29	2500.00	5000.00	2014-09-27	4750.00	1015	590
251	2014-10-25	25000.00	50000.00	NULL	NULL	NULL	593
252	2014-10-27	250.00	500.00	NULL	NULL	NULL	594
253	2014-10-27	250.00	500.00	NULL	NULL	NULL	595
254	2014-10-27	250.00	500.00	NULL	NULL	NULL	596
1103	2014-11-04	30000.00	45000.00	2015-01-21	42500.00	1000	590
1104	2015-12-04	30000.00	45000.00	NULL	NULL	NULL	590

Trans table

Note: For rows in Trans table when customerID is NULL, it means that the art works have not been sold off yet.

Figure Q3(c)

Formulate SQL queries to perform the following tasks:

Question 3c(i)

List the works where the description does not contain the word **expressionist** but the title should have the word **blue**. List in ascending order of artistID.

WorkID	Title	Copy	Medium	Description	ArtistID
590	Blue Interior	Unique	Tempera on card	43.9 x 28 in.	17
571	Yellow Covers Blue	Unique	Oil and collage	71 x 78 in. - Signed	18

(4 marks)

Question 3c(ii)

List the artists with work pieces that are currently not sold yet. Use subquery without any join.

ArtistID	LastName	FirstName	Nationality	DateOfBirth	DateDeceased
11	Sargent	John Singer	United States	1856	1925
17	Tobey	Mark	United States	1890	1976
18	Horiuchi	Paul	United States	1906	1999
19	Graves	Morris	United States	1920	2001

(6 marks)

Question 3c(iii)

List the artists and the number of times their works were sold off and the number of works currently unsold. Do not consider artists not in the Trans table.

artistID	lastName	firstName	nationality	sold	unsold
5	Chagall	Marc	French	1	0
11	Sargent	John Singer	United States	0	1
17	Tobey	Mark	United States	6	2
18	Horiuchi	Paul	United States	1	1
19	Graves	Morris	United States	0	5

(5 marks)

Question 4

Demonstrate the basic techniques of concurrency control and their application.

Consider the tables in Figure Q4(a).

Question 4a

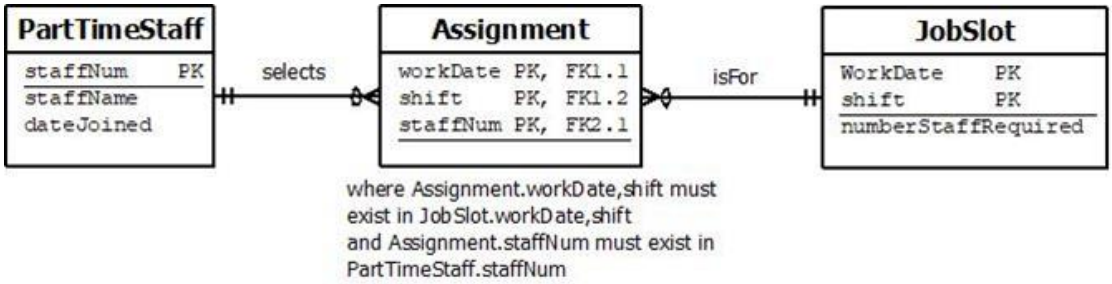

Figure Q4(a)

Suppose part time staff are selecting a work date and shift while concurrently, an administrative staff is updating the number of staff required for job slots.

To help the part time staff select an available work date and shift, the system reads the number of staff required for each job slot and then count the number of assignments rows for the slot and finally, indicates whether the slot is available. After the part time staff has selected an available slot, an assignment row is inserted to record his choice.

Before the administrative staff updates the number of staff required for job slots, the system counts the number of assignments for a job slot and indicates the count against each job slot. After the administrative staff makes the update, if there are assignments affected by the update, the system sends a notification to affected part time staff to inform the staff that his choice has been removed. The removal is based on the last in first out basis, meaning the later assignments are removed first if assignment rows must be removed because of the updates to the number of staff required for job slots.

Discuss a problem that can arise when a wrong isolation level is used. You should state the isolation level that leads to the problem discussed as well as to explain how the problem can arise.

Then propose an isolation level for running the transaction for either the administrative staff or the part time staff such that the problem you have discussed will not happen.

Explain why the proposed isolation level cannot cause the problem to happen.

(6 marks)

Question 4b(i)

Describe **ONE (1)** permission that should be given to part time staff using the database for Question 4(a), based on the tables in Figure Q4(a).

(2 marks)

Question 4b(ii)

Describe **ONE (1)** permission that should not be given to part time staff. Specify the permissions in terms of these actions: select, delete, insert and update on user tables in Figure Q4(a) only.

(2 marks)

Question 4c

Given Figure 4(c), explain what the database recovery procedure will do for the transactions T1 to T3. Explain also how the transaction log help ensure the ACID properties of T1 to T3.

Record Pointer	Transaction ID	Reverse Pointer	Forward Pointer	Time	Type of Operation	Table	Row ID	Before Image	After Image
1	T1	0	2	11:01:00	Start				
2	T1	1	3	11:01:36	Update	Table1	2	2, 12, …	2, 16, …
3	T1	2	9	11:02:55	Update	Table2	14	14, 2, 7, …	14, 2, 5, …
4	T2	0	5	11:03:01	Start				
5	T2	4	8	11:03:24	Update	Table3	15	15, 3, …	15,5, …
6	T3	0	7	11:04:41	Start				
7	T3	6	0	11:05:05	Insert	Table1	3		2, …
8	T2	5	10	11:05:42	Delete	Table3	12	12, 6. …	
Checkpoint									
9	T1	3	0	11:15:00	Rollback				
10	T2	8	11	11:17:57	Insert	Table2	17		17, 5, …
11	T2	10	0	11:19:06	Insert	Table2	18		18, 4, …
Crash!									

Figure 4(c)

(10 marks)

----- END OF PAPER -----